Population metagenomics of a symbiont across its ecologically divergent Tephritid fly host ecotypes



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Abstract

Bacterial symbionts are central to its host ecology and evolution, yet little is known about their evolutionary dynamics during the earliest stages of host divergence. The tephritid fly *Tephritis conura* has recently split into two host plant-associated ecotypes, providing a natural system to test whether their extracellular gut symbiont, *Candidatus Stammerula tephritidis* (*Stammerula*), also has codiverged with their host. To investigate the potential role and evolutionary patterns of divergence in the symbiont, I assembled and annotated the metagenome-assembled genome of *Stammerula* by mining out microbial data from whole-metagenome sequences. The reconstructed 1.48MB genome was nearly complete and highly contiguous with clear signs of genome reduction. Functional analysis revealed that *Stammerula* has the complete pathways to provide essential nutrients to its host. *Stammerula* has the complete pathways for nine of the ten essential amino acids for insects and two B-vitamins (riboflavin and folate). Population genomic analysis of the metagenome-assembled genome showed that *Stammerula* lineages are broadly structured in congruence with their host ecotypes, reflecting ecological divergence but also horizontal transfers between host populations, transects and contact zones, as population differentiation was not correlated with geographic distance. The horizontal transfers were also evident by the comparison of the host phylogeny and symbiont phylogeny. Together, these results bring the tephritid symbiont into the genomic era, providing first insights into its biology and showing that even in the early stages of host ecological divergence, symbionts can begin to co-diverge with their host, offering a snapshot of insect-microbe coevolution in progress.

Introduction

The past, future and present biodiversity reflects the long-term interactions between organisms and their environments. Among the most ancient and persistent of these interactions are those between multicellular organisms and microbes. From soil and water to the internal tissues of animals and plants, life is not only surrounded by microbes, it is permeated by them. Just as microbes are key players for the biogeochemical nutrient cycles of the world, these microbial residents are not passive bystanders.

The community of microbes living within other organisms, the microbiome, can influence nutrition and development (Ben-Yosef et al., 2010; Coon et al., 2016; Elgart et al., 2016; Fitt & O'Brien, 1985; Sannino et al., 2018; Shin et al., 2011), reproduction (Damodaram et al., 2016; Hamden et al., 2013; Noman et al., 2021), behavior (Jia et al., 2021; Jose et al., 2019; Lizé et al., 2014; Morimoto et al., 2019; Schretter et al., 2018; Wong et al., 2017), immunity (Lee et al., 2013), survival and longevity (Akami et al., 2019; Ben-Yosef et al., 2008; Brummel et al., 2004; Khaeso et al., 2018; Piper et al., 2017), as well as resistance to environmental stressors such as insecticides (Cheng et al., 2017). These effects on host development, physiology, and ecology form the cornerstone of the holobiont concept (Margulis & Fester, 1991), which views an organism (biont) and the harmful or harmless microbes living on and within it as a collective evolutionary unit (Zilber-Rosenberg & Rosenberg, 2008). The microbiome formed with the microbiota and their host is an active and important component of the host's biology and adaptation to its environment (Berg et al., 2020; Zilber-Rosenberg & Rosenberg, 2008).

The importance of microbes to the ecology and evolution of their hosts is exemplified in a range of remarkable insect-microbe symbioses in which both host and symbiotic microbes,

symbionts, have coevolved a considerable dependency upon each other (Cornwallis et al., 2023; Douglas, 1989). In these relationships, the symbiont provides a reliable biological function that would otherwise be costly or unavailable to the host, creating a selection pressure on the host favoring mechanisms that retain and pass on the symbiont to their offspring. In return, the symbiont gains a stable and resource-rich environment in which to thrive, released from the pressures of a free-living microbe. Thus, various levels of partnerships, from vital and strictly dependent to opportunistic and flexible, can arise from this symbiotic coevolutionary spiral. Symbionts are therefore usually split into two main symbiont categories: (1) obligate symbionts, where both host and symbiont are mutually dependent on each other for survival, and (2) facultative symbionts, where the host is not dependent on the symbiont for survival, but the presence of the microbe influences fitness of the host (Engel & Moran, 2013; Moran et al., 2008). What differentiates these categories is that obligate symbionts are strictly vertically transmitted, mutualistic, and incapable of surviving outside the host, while facultative symbionts may be either mutualistic or parasitic, transmitted vertically or horizontally, and do not necessarily rely on the host for survival (Kucuk, 2020; Masson & Lemaitre, 2020).

On one hand, mutualistic symbiosis offers clear advantages for both the host and the symbiont. On the other hand, symbiosis has morphological, behavioral and molecular consequences for both partners. To maintain a symbiont, the host has to adapt its immune system to recognize the symbiont as harmless (Gerardo et al., 2010; Ratzka et al., 2013; Wang et al., 2009), develop specialized structures or cells to house the symbiont, and evolve mechanisms and behaviors to reliably pass on the symbiont to the next generation (Bennett & Moran, 2015; Engel & Moran, 2013). For the symbiont, the transition from a free-living microbe to a stable obligate symbiont is an evolutionary “rabbit hole” that leads the microbe down an irreversible path toward extreme specialization and dependency (Bennett & Moran, 2015). The strict vertical transmission of the isolated symbiont lineages causes repeated population bottlenecks, greater genetic drift, and reduced efficacy of selection. As there is no selection to maintain cellular processes necessary for survival outside the host, the symbiont typically develops a characteristic streamlined, AT-biased genome with fewer functional genes (Bennett & Moran, 2015; Perreau & Moran, 2022; Wernegreen, 2002).

This coevolutionary process therefore leads the host and symbiont to diverge together, evident by the fact that host and symbiont phylogenies often mirror each other (co-cladogenesis or phylogenetic congruence) (Moran & Telang, 1998; Page & Charleston, 1998). However, few studies have examined the evolutionary dynamics of symbionts during early stages of host divergence. Does the population structure of symbionts in young evolutionary lineages have phylogenies reflecting those of the hosts? Are there genomic regions under divergent selection? Here, I address these questions by investigating the population genomics of a symbiont whose fly host has recently evolved into two divergent host plant specialist lineages.

Tephritid flies are true fruit flies (Diptera: Tephritidae) whose larvae feed on various plant tissues, including fruits, seeds, flowers, buds, stems and roots. Most members of the Tephritidae family are specialists on a narrow range of host plants (Headrick & Goeden, 1998). As many are also agricultural pests, considerable effort has been devoted to understanding the mechanisms that allow Tephritid flies to colonize and exploit new hosts plants (Shi et al., 2022). Although not an agricultural pest, *Tephritis conura* is a specialist with a history of host plant shifts. Adult flies oviposit their eggs into the flower buds of thistles in the genus *Cirsium*, on which the larvae feed until pupation. During oviposition, eggs are smeared with bacteria that are harbored extracellularly in specialized cell structures in the fly’s mid gut called crypts (Petri, 1909; Stammer, 1929). Mazzon et al (2008, 2010) analyzed the 16S rRNA sequences

of these crypt-associated bacteria in *Tephritis* flies and found that they belong to a novel lineage of Gammaproteobacteria that they called *Candidatus Stammerula tephritidis* (henceforth *Stammerula*). *Stammerula* was found to be unculturable in the lab and exhibited significant but weak co-cladogenesis with its host species (Mazzon et al., 2008, 2010). Together with the fact that *Stammerula* is maternally transmitted during oviposition, these findings suggest that this symbiont plays an important, but unknown part for *T. conura*. To uncover this unknown role, its genome was assembled and annotated to gain the first functional insights into this symbiont.

In continental Europe, *T. conura* has adaptively diverged into two distinct host races, or host-associated insect ecotypes exhibiting significant reproductive isolation with some persistent gene flow (Drès & Mallet, 2002). The ancestral *C. heterophyllum* (CH) specialists and the derived *C. oleraceum* (CO) specialists can be found both in allopatry (geographically separated) and sympatry in northern Europe and Scandinavia (Diegisser et al., 2006, 2007, 2008; Seitz & Komma, 1984; Steward et al., 2024). In the context of the recent adaptive divergence of these fly host races, *Stammerula* presents a compelling opportunity to investigate the evolutionary processes of a symbiont, with the potential to provide novel insights into symbiont adaptation in the early stages of ecological speciation.

In this project, I mined microbial data from whole-metagenome sequence libraries from adult male *T. conura* flies sampled from different locations and host races. Using this subset, I reconstructed and annotated the genome of *Stammerula* by assembling and grouping the metagenome into metagenome-assembled genomes (MAGs) and identified the *Stammerula* MAG using the previously published 16S rRNA sequences (Mazzon et al., 2008, 2010). This approach allowed me to recover and functionally annotate the 1.48 mega base pair (MB) high-quality MAG of *Stammerula*, providing the first functional insights into this symbiont. Using the MAG as a reference for population level genomics analysis, I investigated genetic diversity and divergence of *Stammerula* across host race populations. I found considerable population structure in *Stammerula*, showing this structure was better explained by fly host race than by geographic region or distance between samples. Together these results show that *Stammerula* is dynamically evolving with its fly hosts.

Methods

Insect Samples and DNA Preparation

Microbial sequences were mined from whole metagenome sequence data previously collected from adult male flies. Steward et al. (2024) sampled flies that emerged from CH and CO thistle buds collected from eight locations in northern Europe and Scandinavia to investigate the genomic basis of adaptation to different thistle hosts in *T. conura*. In short, the thistle buds were collected from two sympatric and two allopatric populations for each host plant to enable comparisons between host races as well as between contact zones (Table 1, Figure 1). A single adult male fly was sampled from each bud to avoid including siblings. Each sampled fly was ground with a plastic pestle and DNA was isolated using the Qiagen DNeasy Blood and Tissue kit. Sequencing libraries were prepared with Illumina TruSeq PCR-free kit (350 bp insert size), and whole genome sequencing was performed on Illumina HiSeqX platform (2 x 150 bp paired-end reads) and NovaSeq6000. A single pupa of unknown sex was selected from the CHST population for long read, PacBio sequencing carried out by SciLifeLabs using

their in-house PacBio protocol. See Steward et al. (2024) for details about extraction and sequencing.

Table 1. Description of corresponding metadata for each sample that was sequenced with Illumina. Total samples passed are the number of samples used for the population genomics analysis.

Population ID	Host Plant	Transect	Contact zone	Location	Lat	Long	*Total samples*	*Total samples passed*
CHST	*C. heterophyllum*	West	Allopatric	Sweden	59.628	14.577	9	6
CHSK	*C. heterophyllum*	West	Sympatric	Scania	55.864	13.746	12	11
COSK	*C. oleraceum*	West	Sympatric	Scania	55.905	13.413	12	10
COGE	*C. oleraceum*	West	Allopatric	Germany	51.113	9.597	11	6
CHFI	*C. heterophyllum*	East	Allopatric	Finland	61.352	22.056	9	9
CHES	*C. heterophyllum*	East	Sympatric	Estonia	58.436	22.938	10	7
COES	*C. oleraceum*	East	Sympatric	Estonia	57.879	26.249	10	7
COLI	*C. oleraceum*	East	Allopatric	Lithuania	55.652	21.938	11	11

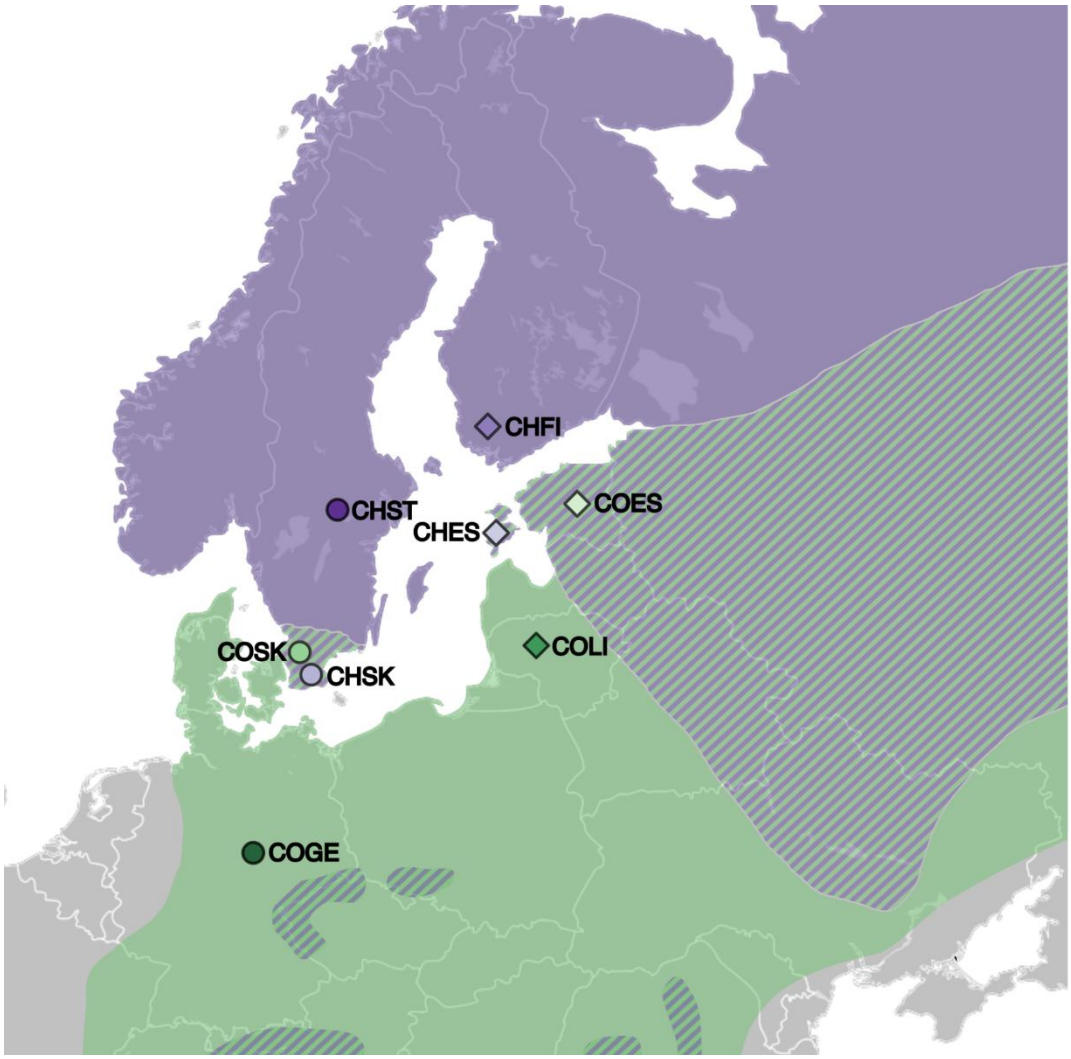


Figure 1. Geographical distribution of sampled *Tephritis conura* host race populations. Transects are represented by circles (west) and diamonds (east); Purple points represent the *Cirsium heterophyllum* (CH) host race whereas *C. oleraceum* host races (CO) are represented by green points. The underlying map is shaded by the distribution of the *Cirsium* plants, with the allopatric ranges of *C. heterophyllum* in purple and *C. oleraceum* in green. Striped areas show contact zones where the plants are sympatric.

Obtaining the *Stammerula* Metagenome-Assembled Genome (MAG)

I used the PacBio long read sequences to assemble the metagenome of *T. conura*. No trimming was done on the long reads as the FastQC v0.12.1 (Andrews, 2010) reports confirmed that the read quality was high (mean Phred: 88, mean length: 17809, total reads 4.2 million). To decontaminate the reads from *T. conura* and human reads, I competitively aligned the reads competitively using minimap2 v2.28 (Li, 2018, 2021) against a 14.52 giga base pairs (GB) concatenated reference FASTA comprising the *T. conura* host genome, the human genome (GRCh38.p14), and 2387 bacterial genomes (Supplementary Table 1A). All reads that aligned to the host or human regions of the competitive reference with an alignment quality above 20 were discarded from the BAM file. I reconstructed FASTQ files from the remaining alignments to retain 174,184 decontaminated FASTQ reads (avg. length: 17514 bp) using bedtools v2.31.1 (Quinlan, 2014). Because the competitive reference contains multiple genomes, a single read can align to multiple locations. When reconstructing FASTQ files from BAM, each alignment is treated as a separate read, which artificially creates duplicates. To correct for this, we used, seqkit rmdup v2.10 (Shen et al., 2024) with the –by-name flag to remove these synthetic copies.

Metagenome Assembly and MAG Quality Assessment

The decontaminated long reads were then assembled with metaMDBG v1.1 (Benoit et al., 2024), creating a 115 mega base pair (MB) metagenome consisting of 2750 contigs derived from the various microbiota of the sample. In metagenomics, binning refers to the process of separating the metagenome into distinct genomes using programs that clusters contigs based on sequence coverage, sequence characteristics (tetra-nucleotide frequency, GC content), or marker gene phylogenies and abundance patterns (Mande et al., 2012). These groups of contigs (bins) represent potential complete genomes. However, these bins are often fragmented, incomplete, or contaminated with reads from closely related taxa. I therefore followed the MIMAG (Minimum Information about a Metagenome-Assembled Genome) standard developed by the Genomic Standards Consortium (GSC; (Bowers et al., 2017)) to assess the quality of the bins and whether they meet the requirements of being MAGs. By the MIMAG definitions, a high-quality MAG must have at least 18 of the 20 amino acid tRNAs genes and the presence of 23S, 16S, and 5S rRNA genes. In addition, contamination must be less than 5% and completeness above 90% where contamination. Medium-quality MAGs are defined by having a completeness score above 50% and contamination below 10%, without the rRNA or tRNA requirements. All other bins below 50% completeness and less than 10% contamination are defined as low-quality MAGs.

To reconstruct MAGs I used metaWRAP v1.1 (Uritskiy et al., 2018) to bin the assembled metagenome using three popular binning tools: CONCOCT v1.1.0 (Alneberg et al., 2014), MaxBin2 v2.2.7 (Wu et al., 2016), and MetaBAT2 v2.15 (Kang et al., 2019). The decontaminated Illumina short read data (described below) from the CHST population was used together with the metagenome in metaWRAP to estimate coverage of each contig, as contigs from the same genome are expected to have similar coverage. The resulting bins from each program were then run through metaWRAP's bin_refinement module to find the best consensus bins with a completeness threshold above 50% and contamination threshold below

10% to only keep medium- and high-quality MAGs. Because metaWRAP calculates completeness and contamination using an outdated version of CheckM (Parks et al., 2015), each of these refined MAGs were assessed with CheckM2 v1.1.0 (Chklovski et al., 2023). Completeness and contamination were quantified using the CheckM2 database v1.1.0 and BUSCO v6.0.0 (Manni et al., 2021) with the bacteria_odb10 database (Kriventseva et al., 2019). Furthermore, gene calling was done using Bakta v1.11.3 (Schwengers et al., 2021), and each MAG was taxonomically classified using GTDB-Tk v2.4.0 (Chaumeil et al., 2022) with the r220 GTDBK-Tk reference database. Standard assembly metrics of each MAG were calculated using the stats.sh script from BBmap v39.13 (Bushnell, 2014).

To identify the MAG belonging to *Stammerula*, I used BLAST+ v2.16.0 (Camacho et al., 2009) to create a custom database containing the refined MAGs from metaWRAP together with the curated and non-redundant small subunit (SSU) SILVA reference database (SILVA_138.2_SSURef_NR99) (Quast et al., 2013; Yilmaz et al., 2014). The SILVA database contains 510,508 curated SSU sequences, including *Stammerula* 16S rRNA records from other tephritid flies such as *Phaeogramma vittipennis* (KM079240.1), *Trupanea arboreae* (KM079241.1), *Oxyna flavipennis* (EF469630.1), *Trupanea amoena* (EF469626.1), *Acanthiophilus helianthi* (EF469625.1), *Campiglossa doronici* (EF469636.1), *Sphenella marginata* (EF469629.1), and *T. arnicae* (EF469616.1). I queried the custom BLAST database with previously published *Stammerula* 16S rRNA sequences isolated from *T. cometa* (EF469615.1), *T. hyoscyami* (EF469613.1), and *T. conura* (EF469618.1; (Mazzon et al., 2008, 2010). The bin containing high identity hits to these 16S sequences was considered to be the Stammerula MAG.

Functional Annotation

I assessed the potential functional profile of *Stammerula* by analyzing KEGG pathways encoded in the *Stammerula* MAG. Genes were initially annotated with Bakta, but because Bakta does not assign KEGG orthologs (KOs), the predicted protein sequences were further annotated using KEGG's BlastKOALA (Kanehisa et al., 2016). To broaden the annotation and reduce the limitations of a single tool (Supplementary Figure 1), the *Stammerula* MAG was also annotated with DRAM v1.5.0 (Shaffer et al., 2020) and Mettannotator v1.4.0 (Gurbich et al., 2025). KO annotations from all three pipelines were merged into a non-redundant union set, which was then assessed for pathway completeness with MGnify's kegg-pathways-completeness tool v1.3.0 (Richardson et al., 2023). Finally, the union set was manually cross-checked in KEGG Mapper V5 (Kanehisa et al., 2022; Kanehisa & Sato, 2020) to recover potential missed pathways. All archaea, plant, and animal pathways were excluded, resulting in completeness scores for 169 KEGG pathways.

Pipeline for up-stream metagenomics and MAG assessment

Metagenomics is a smörgåsbord of programs and workflows with a multitude of outputs to consider for complete analysis. To automate metagenome assembly, bin refinement and required outputs needed for MAG evaluation, I built a Nextflow pipeline that handles the up-stream analyses to give the required outputs for assessing the produced MAGs for downstream analyses such as population genomic analysis. The pipeline automates metagenome assembly, binning, gene calling, MAG quality, and taxonomy classification, and was designed to accommodate both short and long reads, and their individual sample structure. By structuring the workflow with Nextflow, the pipeline ensures reproducibility, scalability and portability as it is created using containers. This pipeline streamlined the

analysis of *Stammerula* but is also built to handle other host-microbiome systems using both long and short reads. Instructions and code are available at my GitHub repository: https://github.com/ndreey/stammerula2025.

Population Genomic Analyses

To evaluate how *Stammerula* varies across host races, regions and contact zones, I performed a population genomic analysis. The analysis focused on the *Stammerula* MAG recovered from long read metagenome assembly and included variant calling, filtering and multiple layers of population genomic inferences. The steps included aligning decontaminated reads to the metagenome, generating and filtering a high-quality variant dataset, and performing statistical analyses assessing genetic structure, phylogenetic reconstruction, and estimates of differentiation and diversity. Together, these analyses aim to uncover symbiont patterns of population structure and identify candidate genomic regions linked to the ecologically divergent environments experienced by the specialist host fly lineages.

Pre-processing of short reads

I trimmed the Illumina whole-metagenome sequence library (short reads) using fastp v0.23.4 (Chen, 2023) to remove adapters, duplicates, and low-quality reads (Phred score < 20; length < 36 bp). The quality of the reads pre- and post-trimming was assessed by summarizing the FastQC v0.12.1 and fastp reports using MultiQC v1.12 (Ewels et al., 2016) (Average length: 149.6, 95% of the reads passed trimming, average Phred score > 30). As with the long reads used to assemble the metagenome and *Stammerula* MAG (see above), I decontaminated the short reads by aligning reads against the concatenated competitive reference (Supplementary Table 1A). The short reads were aligned with BWA-MEM v0.7.17 (Li, 2013) and I discarded all read pairs where either mate mapped to host or human regions before reconstructing paired FASTQ files using SAMtools v1.20 (Danecek et al., 2021) for the paired end reads. Lastly, for samples sequenced on multiple lanes, the corresponding reads for each sample were merged. Proper pairing of the decontaminated, merged FASTQ files was assessed and potential artificial duplicates from the decontamination process were removed with seqkit v2.10.0.

Variant Calling

To understand how *Stammerula* differs between and within populations, I called variants by aligning the decontaminated short reads from each sample against the full metagenome using BWA-MEM. Aligning to the entire metagenome, instead of the *Stammerula* MAG alone, avoids forcing reads from related taxa onto the *Stammerula* MAG and thus minimizes false positives. The resulting BAM files were sorted with SAMtools and PCR duplicates were marked with MarkDuplicates v3.3.0 (*Picard Toolkit*, 2019)**.** Variants were then called with ploidy set to 1 using GATK's HaplotypeCaller v4.5.0.0 (Poplin et al., 2018) to generate individual GVCFs, which were combined with GATK CombineGVCFs and genotyped jointly using GenotypeGVCFs with the –all-sites flag to retain both variant and invariant sites. Finally, to only keep records corresponding to the *Stammerula* MAG, I used BCFtools v1.20 (Danecek et al., 2021) to subset the VCF to the contig comprising the *Stammerula* MAG.

VCF filtering

After subsetting to the contig of the *Stammerula* MAG, I processed the raw VCF file to generate a high-quality, filtered variant dataset suitable for downstream population genomic analysis. This workflow began by separating variant and invariant sites using BCFtools. I extracted all

biallelic SNPs and applied variant hard filtering following GATK best-practices to minimize false positives (*Hard-Filtering Germline Short Variants*, 2024). These best-practices include filtering on depth-normalized quality (QD), strand bias using FisherStrand (FS) and StrandOddsRatio (SOR), and finally the variation of mapping quality at each site using RMSMappingQuality (MQ). Using GATK VariantFiltration v4.5.0.0 I flagged variants to discard within these thresholds: QD < 2.0, SOR > 3.0, FS > 60.0, and MQ < 40.0 and kept only the ones that passed using BCFtools. To further improve variant quality, I applied proximity filtering (Bush et al., 2020) by removing SNPs located within three base pairs of another SNP using VCFtools v0.1.16 (Danecek et al., 2011) and SNPs within 10 base pairs of an indel using BCFtools. No filtering was done on the invariant sites. Lastly, I merged the filtered biallelic records with the invariants to produce two final VCF files: the filtered biallelic SNP VCF and the merged VCF with filtered biallelic SNPs and invariants sites. The final VCF files were validated using GATK ValidateVariants and BCFtools stats. I evaluated variant quality by summarizing site-level statistics (QUAL, DP, FS, SOR, MQ, QD, MQRankSum, ReadPosRankSum, AN, AC, AF) with BCFtools and calculating both site- and individual-level missingness using VCFtools.

Coverage and missingness of the *Stammerula* MAG

I estimated coverage using Qualimap multi-bamqc v2.3 (Okonechnikov et al., 2016) on the metagenome alignment files but restricted multi-bamqc to the *Stammerula* MAG using the –feature-file option with a BED file converted from the .gff3 Bakta annotation. Read counts, number of aligned reads, mean coverage, and GC content were extracted from the Qualimap reports using a custom parser. Correlations between individual-level missingness and the Qualimap statistics were evaluated in R v4.3.3 (R Core Team, 2024). Following this assessment, 17 samples were removed from both VCF files based on thresholds of missingness > 0.7 and log10(aligned reads) < 3.8.

Population structure

Population structure was assessed using principal components analysis (PCA) implemented in PLINK v1.90b7.7 (Purcell et al., 2007) on the filtered biallelic SNP VCF. Principal components were visualized in R. One sample (CHSK_P14052_101) was identified as an outlier and removed before repeating the PCA. The resulting binary BED file was then used for ancestry analysis with ADMIXTURE v1.3.0 (Alexander et al., 2009). PCA and ADMIXTURE provide complementary perspectives on genetic structure as PCA summarizes major axes of genetic differentiation, while ADMIXTURE estimates individual ancestry proportions based on number of clusters (K). The number of clusters were determined by plotting the cross-validation errors and looking for local minimum.

To reconstruct phylogenetic relationships among the samples, the filtered biallelic SNP VCF was converted to a FASTA alignment using vcf2phylip v2.9 (Ortiz, 2019). A maximum-likelihood tree was then inferred with IQ-TREE2 v2.4.0 (Minh et al., 2020), with ModelFinder (Kalyaanamoorthy et al., 2017) enabled to select the best fit substitution model and 500 classic bootstraps replicates (--boot flag) to estimate branch support. The consensus tree was plotted in ggtree (Yu et al., 2017). Host and symbiont phylogenies were compared using a cophylogeny with links between corresponding host-symbiont samples using ape v5.8 (Paradis & Schliep, 2019) and phytools v2.4 (Revell, 2024). Both trees were pruned to shared samples and branch lengths were ignored. The phylogeny of the *T. conura* fly samples,

originally constructed in astral using Dipteran single copy orthologs (Steward et al. 2024), was provided by R Steward (unpublished data).

Genetic diversity and population differentiation

Nucleotide diversity (π) and departures from neutrality (Tajima's D) were calculated for *Stammerula* in each fly population using pixy v2.0.0.beta2 (Korunes & Samuk, 2021) on the merged VCF file containing filtered biallelic SNPs and invariant sites. Population differentiation (Hudson's F_{ST};(Bhatia et al., 2013)) and nucleotide divergence (D_{XY}) were calculated in non-redundant pairwise comparisons among all host race populations. F_{ST} and D_{XY} were calculated over non-overlapping 1kb windows across the genome using pixy, while whole genome F_{ST} and D_{XY} were calculated with piawka v0.8.11 (Scott et al., 2024). To asses population structure, I visualized symmetric matrices of genome-wide pairwise F_{ST} and D_{XY} as heatmaps with pheatmap v1.0 (Kolde, 2025). Rows and column were hierarchically clustered using UPGMA on Euclidean distances derived from the F_{ST} and D_{XY} matrices, and the resulting dendrograms were visualized in the heatmap. The topology was compared against the *T. conura* population topology (Steward et al. 2024). Correlation between geographic distance and F_{ST} was calculated using Pearsons's correlation implemented in R.

Taxon assignment with Kraken2

To assess the decontamination process, and whether *Stammerula* missingness was due to fewer microbial reads or higher abundance of another microbe, I assessed the taxonomic compositions of reads in each decontaminated metagenomic sample using Kraken2 (Wood et al., 2019). A custom Kraken2 database was built using the Kraken bacterial library (v2.1.6), *T. conura* reference genome, and the *Stammerula* MAG. All reads were classified against this database using Kraken2 with the –confidence parameter set to 0.5, which reduces false positives as it forces the read to be classified by the taxa that has the majority of the assigned kmers. To calculate the number of classified reads and relative abundance of taxa across the samples, I used KrakenParser v0.6.1 (Popov, 2025).

Results

Highly complete *Stammerula* MAG

By mining microbial reads from the whole pupa PacBio sequence library (CHST population), I retained 0.04% of the long read sequence data. Assembly of these data resulted in a 115 mega base pair (MB) metagenome composed of 2,750 contigs (N50: 305, L50: 60.217 KB). Binning resulted in 93 total bins (MaxBIN2: 21, MetaBAT2: 35, CONCOT: 37) that were then refined to six bins meeting the MIMAG medium-quality thresholds of minimum 50% completeness and max 10% contamination. Of these, GTDB-Tk classified two bins to the species level and the remaining four to genus level (Table 2), spanning the classes Bacteroidia, Actinomycetes, Gammaproteobacteria, and Alphaproteobacteria. As *Stammerula* belongs to the Enterobacteriaceae family within Gammaproteobacteria, bin 3 (*Erwinia spp*., 1 contig) and bin 4 (*Rahnella rivi*, 10 contigs) were the candidate bins representing *Stammerula*.

Using previously published *Stammerula* 16S rRNA sequences (Mazzon et al. 2008) as BLAST search queries against a custom blast+ database comprising the MAGs and the SILVA 16S rRNA collection, I found that bin 3 was highly likely to represent *Stammerula* (Supplementary

Table1B). The *Stammerula* 16S rRNA queries isolated from *Tephritis cometa* (EF469615.1), *Tephritis hyoscyami* (EF469613.1), and *Tephritis conura* (EF469618.1) all had best matches with contig ctg4994 from bin 3 (Figure 1). Bin 3 consists solely of the 1.48 MB long contig ctg4994, which had 7 unique matches above 99.6% percentage identity (evalue: 0.0) against each *Stammerula* 16S rRNA query. Interestingly, the *Stammerula* 16S sequence isolated from *Tephritis hyoscyami*, not *T. conura,* best matched the 16S rRNA sequences on ctg4994. The query from *Tephritis conura* had 2-3 mismatches while *Tephritis hyoscyami* only had 1-2, depending on the 16S rRNA locus. I used the SILVA records, which include eight *Stammerula* species, as a positive control. The best match to these records was with the *Stammerula* 16S rRNA sequence isolated from *Tephritis arnicae* (EF469616.1; 96.7% identity, [5-56] mismatches).

Table 2. High and medium quality bins (putative MAGs) refined from the assembled metagenome using metaWRAP.

Metrics		**bin.1**	**bin.2**	**bin.3**	**bin.4**	**bin.5**	**bin.6**
Contigs		2	60	1	10	4	135
N50		1	15	1	2	2	35
L50		4 611 505	58 686	1 476 797	1 458 117	466 612	42 215
Contig_max		4 611 505	250 096	1 476 797	1 668 259	529 019	115 044
GC		0.36	0.71	0.43	0.53	0.35	0.41
Size		4 629 864	2 719 439	1 476 797	5 678 801	1 248 500	4 294 708
CDS_Dens.		0.90	0.84	0.75	0.87	0.85	0.84
Tot.CDS		4 321	3 041	1 217	5 284	1 289	5 012
23S		4	1	5	7	1	5
16S		4	1	7	7	1	5
5S		4	2	6	8	1	8
rRNA		12	4	18	22	3	18
tRNA		57	32	49	78	34	40
ori		0	1	0	1	1	0
GTDB-Tk		*Epilithonimonas spp.*	*Curtobacterium spp.*	*Erwinia spp*	*Rahnella rivi*	*Wolbachia pipientis*	*Dyadobacter spp.*
CheckM2	**Comp. (%)**	100	57.25	99.99	100	93.69	51.34
	Cont. (%)	0.28	2.16	0.0	0.3	1.63	6.76
BUSCO	**Comp. (C)**	123 (99.2%)	69 (55.6%)	116 (93.5%)	116 (93.5%)	97 (78.2%)	69 (55.6%)
	Complete and single-copy (S)	123 (99.0%)	67 (54.0%)	116 (93.5%)	116 (93.5%)	97 (78.2%)	66 (53.2%)
	Complete and duplicated (D)	0 (0.0%)	2 (1.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (2.4%)
	Fragmented (F)	1 (0.8%)	17 (13.7%)	4 (3.2%)	4 (3.2%)	10 (8.1%)	14 (11.3%)
	Missing (M)	0 (0.0%)	38 (30.6%)	4 (3.2%)	4 (3.2%)	17 (13.7%)	41 (33.1%)

In addition to strong evidence that bin 3 captured *Stammerula* sequences, it also met the MIMAG standards (see methods) for a high-quality MAG (Table 2). CheckM2 reported that bin 3 (hereafter stamMAG) had a completeness of 99.99% and contamination of 0.0%. Similar results were found by BUSCO comparison with the bacteria_odb10, which showed that of the 124 BUSCO markers, 93.5% were complete, 0% were duplicated, and 3.2% were missing, reinforcing that stamMAG is of high completeness and low contamination. The seven 16S rRNA BLAST hits on stamMAG were confirmed by the gene annotation results from Bakta (Figure 1). No origin of replication was found, but stamMAG consists of 1217 CDS, 16 pseudogenes, 159 hypothetical coding genes, 49 tRNAs (encoding 20/20 amino acids), and 18 rRNA genes (six 5S rRNAs, seven 16S rRNAs, and five 23S rRNAs).

Functional annotation of *Stammerula* gene features

To determine the metabolic role of *Stammerula* in its fly host, I evaluated its functional profile from the encoded KEGG pathways from the three annotation tools. DRAM assigned 937 KOs from 1211 predicted proteins, Bakta 954 KOs from 1217 gene calls, and mettannotator assigned 1003 KOs from 1307 predicted genes. Using the union set of 1023 KOs, the stamMAG revealed a combination of pathways that support its own cellular maintenance and core carbon and energy metabolism while also encoding metabolites that could supplement the nutrition of *T. conura*. Broadly, *Stammerula* is a metabolically capable symbiont which has the biosynthetic capacity for multiple amino acids, vitamins, and cofactors.

Central carbon metabolism

Stammerula encodes a nearly complete central carbon metabolism (Figure 3, group D, E). The Embden-Meyerhof glycolysis pathway is 89% complete, lacking only hexokinase, while the pentose phosphate pathway and the TCA cycle are fully present together with a complete respiratory chain, indicating the capacity for aerobic respiration. The phosphate acetyltransferase-acetate kinase (Pta-Ack) pathway is also complete, suggesting the capability for fermentation under low-oxygen conditions that may occur in the crypt environments. In contrast, the uronate pathway is nearly absent, and genes for xylanases, cellulases, and pectinases were not detected, suggesting that *Stammerula* is poorly adapted to metabolize complex plant cell wall sugars. The genome does, however, encode complete de novo purine and nearly complete pyrimidine biosynthesis, ensuring self-sufficiency in nucleotide production for replication and transcription.

Cellular structures

Cell envelope pathways reveal signs of symbiosis. *Stammerula* retains complete fatty acid and phospholipid biosynthesis, as well as precursors for peptidoglycan, consistent with a functional Gram-negative membrane (Figure 3, group D, G, H). Lipopolysaccharide (LPS) biosynthesis is largely present, including lipid-A anchor and core sugar modules (Figure 3, group D, G). However, the O-antigen pathways are only partial or absent (Figure 3, group C, D), and the lack of the O-antigen ligase gene *waaL*, suggests *Stammerula* has a “rough” LPS phenotype (Kobayashi et al., 2011). In addition, components of flagellar assembly were entirely absent. The lack of flagella and reduction of surface glycan reflects adaptation to a stable host environment where movement is not necessary. Moreover, having a “rough” LPS phenotype may reduce immune recognition from the host (Lerouge & Vanderleyden, 2002).

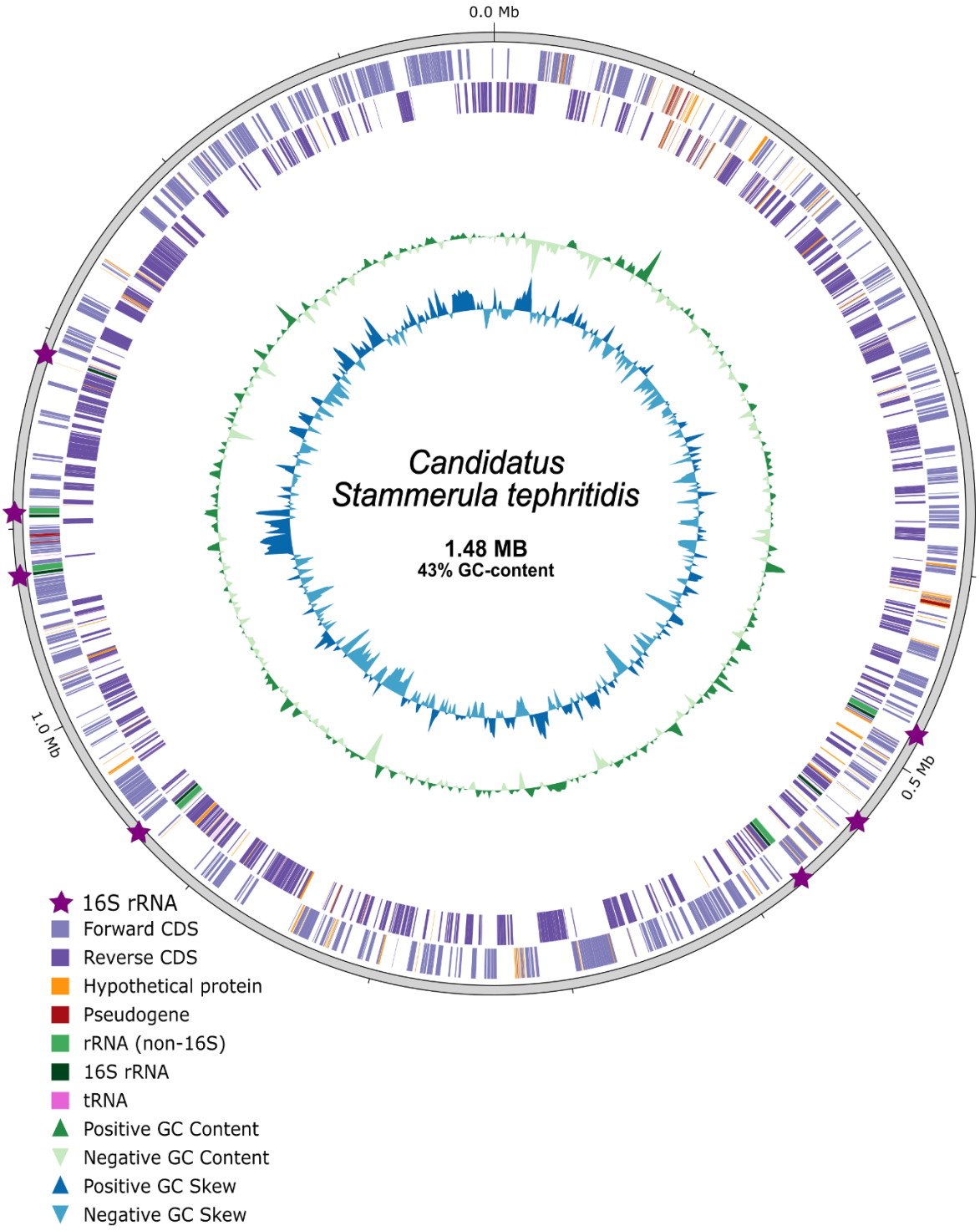


Figure 2. The metagenome-assembled genome (MAG) of the symbiont *Candidatus Stammerula tephritidis* visualized with Bakta annotated features. The 16S rRNA loci are marked in green to show which strand and with stars to quickly find the seven loci that was used to identify that this MAG was the symbiont of *Tephritis conura*. The MAG consists of one single contig and was determined as high-quality following the standard set by the Genomic Standards Consortium.

Essential nutrients

Stammerula has the genetic toolkit for multiple amino acid biosynthesis (Figure 3, group A). *Stammerula* encodes complete pathways for arginine, cysteine, histidine, lysine, methionine, ornithine, threonine, serine, tryptophan, and the branched-chain amino acids (valine, isoleucine, leucine). While *Stammerula* has the complete Shikimate pathway for synthesizing

the precursor for phenylalanine and tyrosine, both of these amino acid pathways are incomplete due to the absence of *tyrB*. In terms of cofactors and vitamins (Figure 3, group I), *Stammerula* displays a mixed repertoire. Complete pathways were identified for riboflavin (B2) and folate (B9), while partial pathways were found for biotin (B7), pantothenate (B5), and pyridoxal phosphate (B6). Thiamine (B1), and cobalamin (B12) are largely absent. Genes for cofactor metabolism are rich, with complete pathways for NAD, coenzyme A, siroheme, lipoic acid, heme, glutathione, and spermidine.

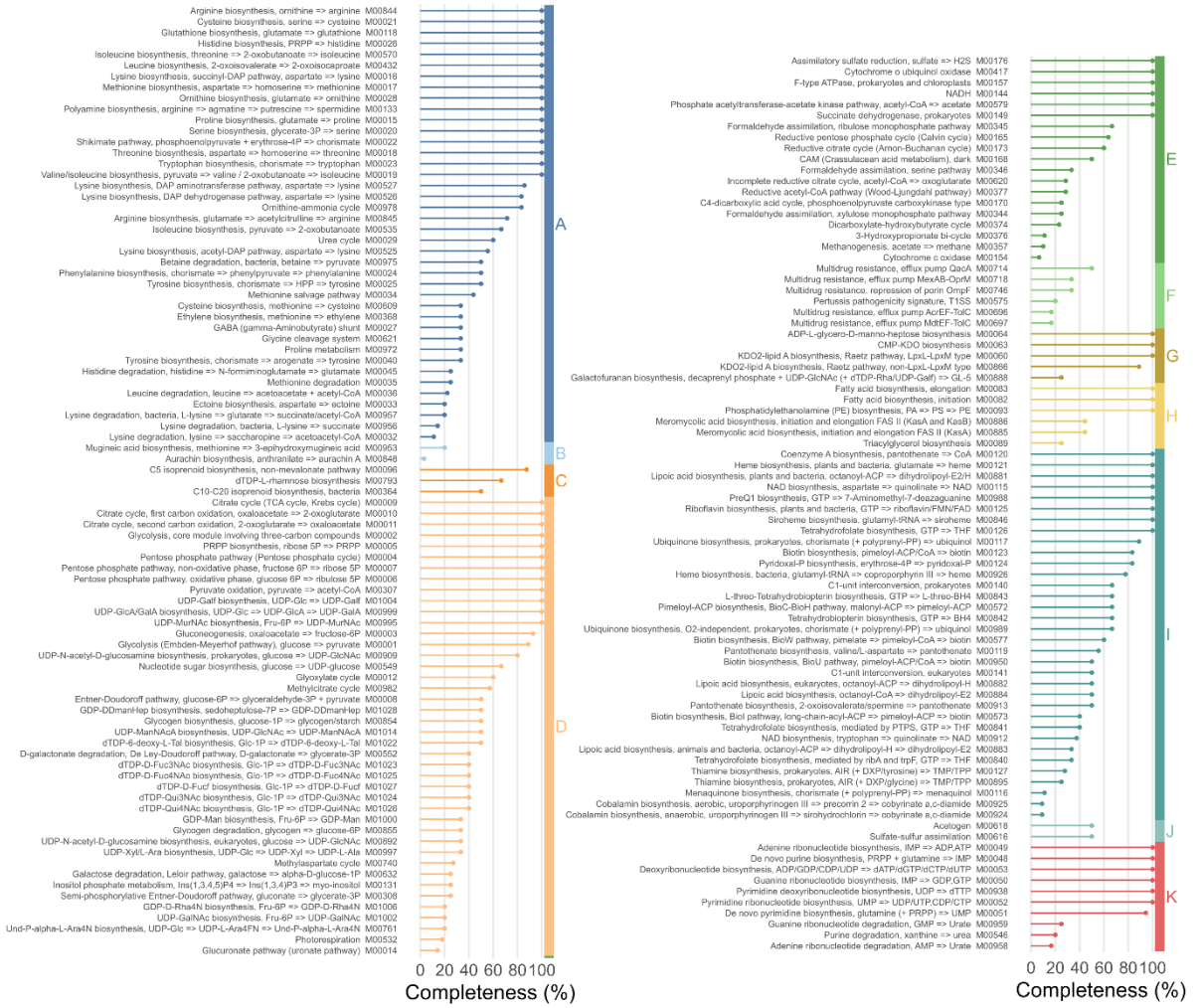


Figure 3. KEGG Pathway Completeness based on presence/absence colored by group: A) Amino acid metabolism, B) Biosynthesis of other secondary metabolites, C) Biosynthesis of terpenoids and polyketides, D) Carbohydrate metabolism, E) Energy Metabolism, F) Gene set, G) Glycan metabolism, H) Lipid metabolism, I) Metabolism of cofactors and vitamins, J) Module set, K) Nucleotide metabolism.

Stammerula was not present in all fly samples

Of the 2.2 billion decontaminated reads, only 0.4% mapped to the stamMAG contig on the metagenome corresponding to an average of ~103,000 reads per sample. These alignments produced 14,281 SNPs and 1,759 INDELs before filtering, and 11,912 high-quality biallelic SNPs after filtering. However, samples from many flies distributed across all populations had high fractions of individual missingness and low mean coverage (Figure 4A, B). Interestingly, missingness was not significantly correlated with the number of decontaminated reads retained in each sample ($r(113) = 0.09$, 95% CI [-0.09,0.27], $p = .31$), i.e., missingness was not simply a function of low sequencing depth (Figure 4B).

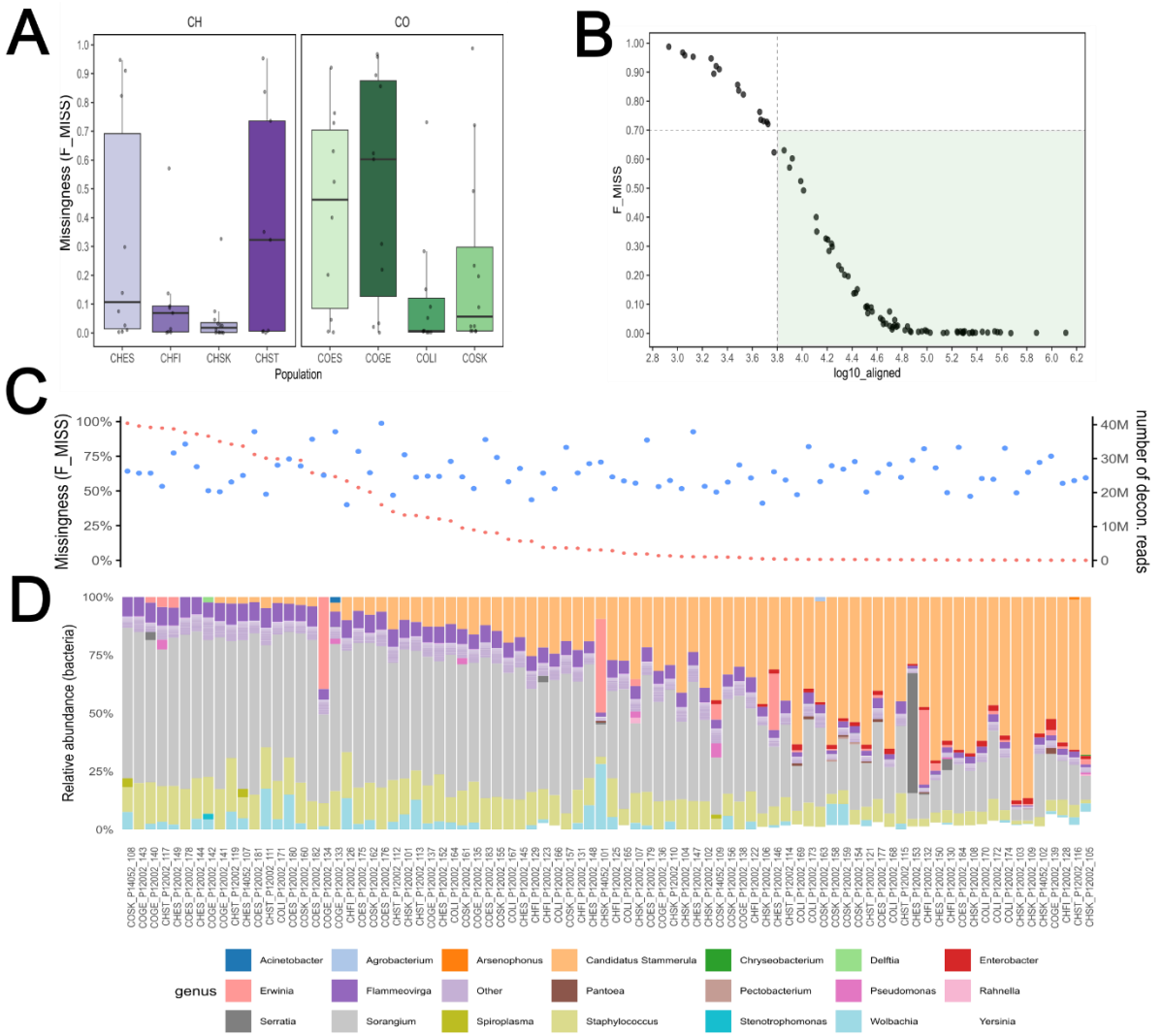


Figure 4. *Stammerula* is not found in all fly samples. A) The distribution of individual missingness per population. Boxes show the interquartile range (IQR) bisected by the median and whiskers extend no further 1.5 * IQR. B) Individual missingness plotted against number of aligned reads against the *Stammerula* metagenome-assembled genome. Green box represents the samples above 3.8 log10(number aligned reads) and less than 0.7 individual missingness. C) Individual missingness (red points) in order from high to low alongside the total decontaminated reads (blue points) used to align against the T. conura metagenome (see B). D) Relative abundance of microbial genera identified by Kraken2, ordered by individual missingness.

To confirm that the apparent absence of *Stammerula* in some samples was biological rather than an artifact of mapping or variant calling, taxonomic classification was done on the decontaminated short reads using Kraken2. Median classification rates were 46.6% for short reads, with 40.2% assigned to *T. conura* and only 0.07-0.08% to *Stammerula* (regardless of taxonomic level). Long reads showed higher classification rates overall (~83.2%), with 78.1% assigned to *T. conura*, and 0.4% to *Stammerula*. This indicates that a large proportion of the reads remained unclassified against the custom Kraken2 database.

The relative abundance calculation from KrakenParser confirmed that the microbial fraction was very low. *Tephritis conura* accounted for a median of 99.3% (across taxonomical levels)

of the classified reads, indicating that the microbial relative abundance from the classified data is less than 1%. Importantly, by only plotting the 1% microbial relative abundance showed that samples with highest missingness were also those with the lowest relative abundance of *Stammerula* in the Kraken2 analysis, they instead had higher representation of *Sorangium* (Family Polyangiaceae, Class Myxococcia; Schneiker et al., 2007) (Figure 4C). This finding supports the conclusion that the missingness reflects biological variation, rather than technical artifacts of the variant calling process.

Stammerula population structure mirrors fly ecotypes

After removing 17 samples with a high fraction of individual missingness (Figure 4D), a total of 67 samples were used in a principal component analysis (PCA). The PCA showed that CHSK_P14052_101 was an outlier and was also removed (Supplementary Figure 2). Repeating the PCA without the outlier showed three clear clusters in the first two PC axes (Figure 5A), two of which grouped samples largely by host fly ecotype (CH or CO) and one group with samples from both fly ecotypes (CH+CO). The CH and CO groups were separated by PC1 which explained 25.3% of the variance while PC2 (14.7% of variance) separated the CH+CO group. This was also supported by the ADMIXTURE results (Figure 5B), for which K=3 and K=6 both showed local minima in the cross-validation error (Supplementary Figure 3), and the phylogeny (Figure 5D, Figure 6).

The maximum-likelihood tree, with a best fit model of K3Pu+F+R2, indicated three highly supported major clades that splits the phylogeny into CH, CH+CO, and CO (Figure 5D, Supplementary Figure 3). This clustering differs from the main pattern of CO diverging from CH, as found in the *T.conura* tree (Figure 6), in which CO and CH host races form monophyletic clades.

Furthermore, the maximum-likelihood tree shows that *Stammerula* is not grouped by population as the population samples are scattered across these three major clades (Supplementary Figure 4). It is only CHSK that shows a tendency of well supported population level clustering as within the CH clade, 8 of the 11 CHSK samples are together. Although poorly supported, all 10 COSK samples are only found within the CO clade. Moreover, structure based on transect or contact zone (allopatry vs. sympatry) can not be found. On one hand, in the CH+CO clade we find one branch with mostly allopatric CH populations but from different transacts. On the other hand, a clear grouping based on east transact can be found in CH+CO, but in this case with mixed contact zones. The only consistency is the clustering of host ecotype and the clear separation between CHSK and COSK that both share the same sympatric contact zone. Lastly, COSK is only found inside the CO clade while the other CO samples are found both in the CH+CO and CH clade.

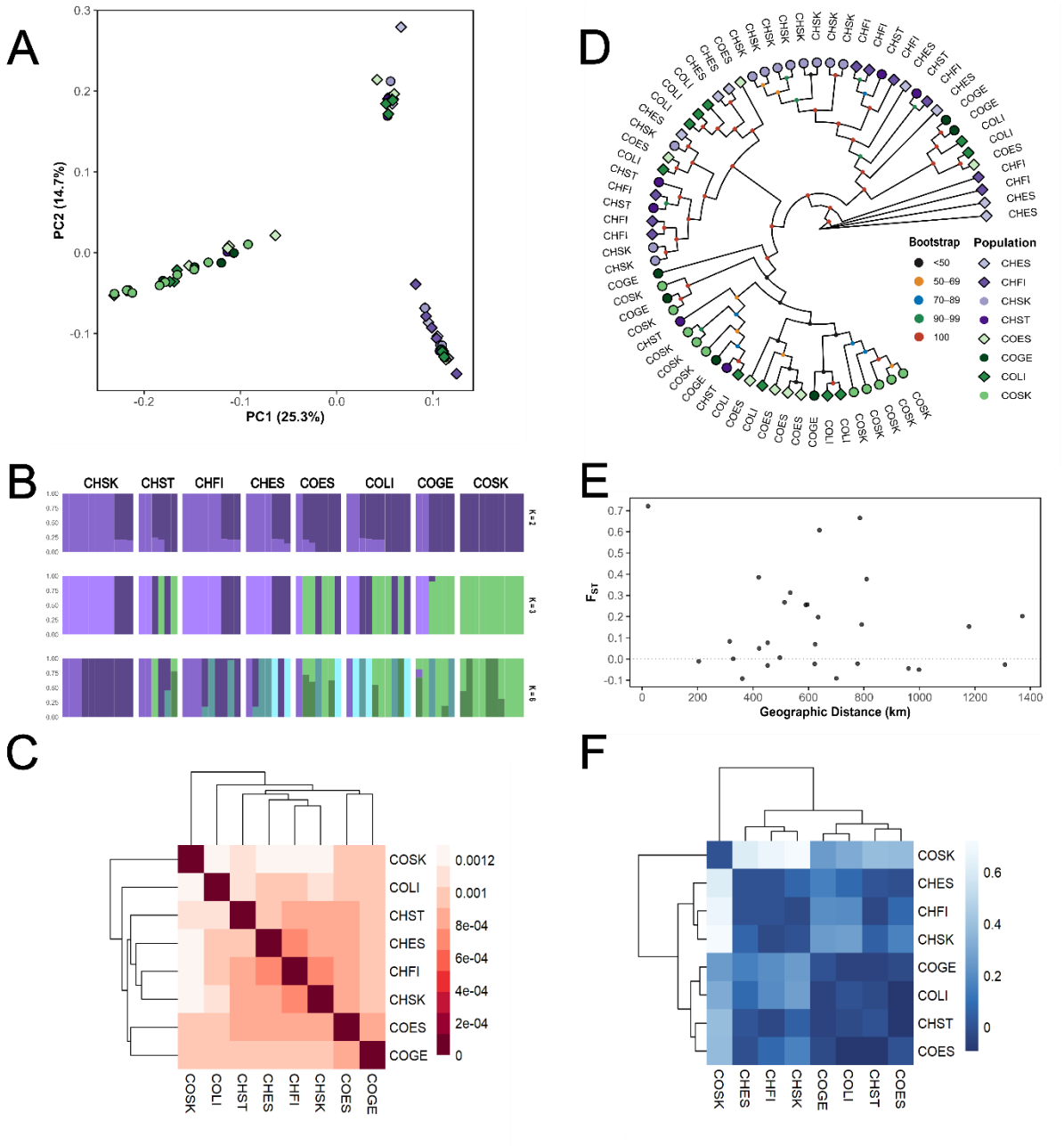


Figure 5. *Stammerula* population structure. A) Principal component plot showing first and second principal component of the filtered biallelic SNPs VCF file excluding outlier individual. B) Admixture proportions of the same biallelic SNPs VCF, estimated for K clusters (2, 3, 6). C. Hierarchical clustering of Euclidean distances from whole genome nucleotide divergence (D_{XY}, red) and (F) population differentiation (F_{ST}, blue) matrices. D. Unrooted maximum-likelihood cladogram derived from the same biallelic SNPs VCF of the 67 samples with 500 standard bootstraps. E. Population differentiation for each non-redundant population pair plotted against geographic distance to show that no isolation by distance.

Co-cladogenesis was visually investigated by comparing the *T. conura* host race ecotype phylogeny with the unrooted maximum-likelihood tree of *Stammerula* by linking the samples together (Figure 6). This tanglegram of topologies shows that the symbiont does not perfectly track its host as *Stammerula* host race populations are scattered across multiple branches whereas the host tree shows a strong population grouping. For example, while COGE forms a monophyletic CO clade in the host tree, its corresponding *Stammerula* symbionts are distributed across the three CH, CH+CO, and CO clades. This pattern suggests that while there is partial phylogenetic congruence, strict co-cladogenesis is absent from this hypothesized *Stammerula* phylogeny. Instead, *Stammerula* appears to be undergoing its own

evolutionary dynamics, loosely tracking host divergence but also showing evidence for independent diversification (Figure 6).

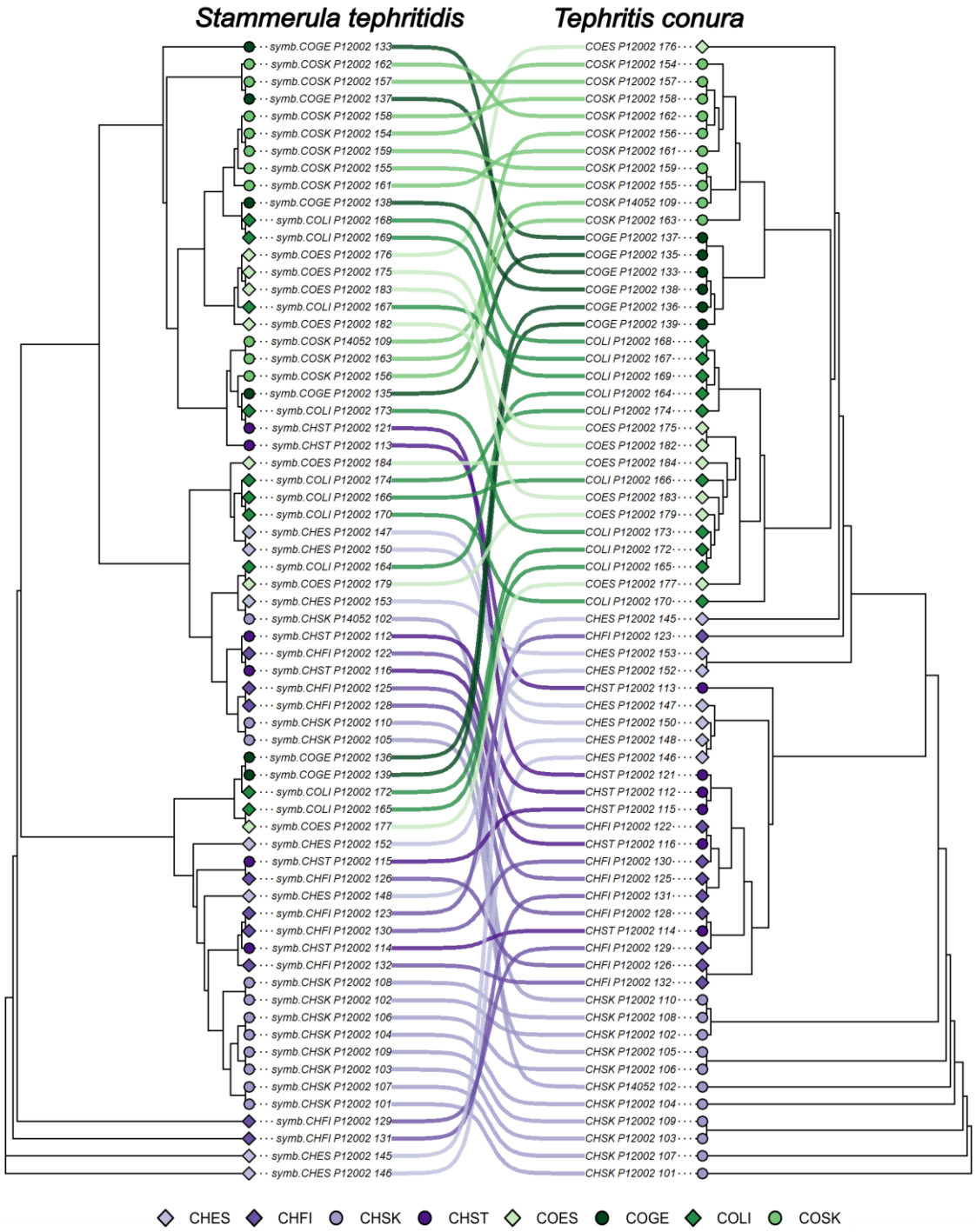


Figure 6. Tanglegram showing the cophylogeny between *Stammerula* (left) and its host *Tephritis conura* (right). The maximum-likelihood tree of *Stammerula* (based on the filtered biallelic SNPs) is plotted against the astral phylogeny of *T. conura* which is derived from single-copy dipterian BUSCOs. Curved links connect mathicn host-symbiont pairs, colored by population identity.

To investigate co-cladogenesis at a lower resolution, I assessed hierarchical clustering of populations based on genome-wide F_{ST} and D_{XY} values. In contrast to *T. conura* flies (Steward et al. 2024), populations did not group by host race for *Stammerula*. Both metrics identified

COSK as the major outlier (Figure 5C, F). Although nonsignificant, a weak negative correlation was found between geographic distance and F_{ST} ($r(26) = -0.14$, 95% CI [-0.49, 0.24], p = 0.46), which is the opposite expectation under isolation by distance (Figure 5E). Removal of COSK from the dataset, changed the correlation toward a weak positive correlation of r(19) = 0.13 with F_{ST} and geographical distance, but remained nonsignificant (95% CI [-.32, 0.53], p = 0.57). Concluding that the genetic differentiation is not accounted for by distance.

Loci of interest

No clear loci of differentiation were identified between CH and CO populations, as F_{ST} was relatively evenly elevated across the genome. Over non-overlapping 1KB windows, mean D_{XY} was 0.0008 (SD = 0.001) and mean F_{ST} was 0.22 (SD = 0.16). Genome-wide Tajima's D averaged -0.41 (SD = 0.8) for CH populations and -0.10 (SD = 0.7) for CO populations, indicating a stronger skew toward rare variants in CO (Figure 7A).

However, the 92-104KB region was potentially of interest as π spiked dramatically (Figure 7B). In CO, π reached 0.024, which is over 20 standard deviations above the mean, and a positive Tajima's D, suggestive of balancing selection and the presence of multiple coexisting haplotypes. In contrast, CH showed lower nucleotide diversity (π = 0.005, ~4 SD above mean) and strong negative Tajima's D (-2), consistent with a selective sweep.

This region encodes a toxin-antitoxin (TA) system, including the *HigB* toxin, its *HigA* antitoxin, and the RelE mRNA interferase, alongside a ISL3 transposase. TA modules are widespread in bacterial chromosomes and have influence virulence factors, swarming motility, biofilm formation, and influence the type III secretion system (T3SS) (Zhang et al., 2018). Factors that could be of importance when colonizing the host crypts (Engel & Moran, 2013). Therefore, the contrasting Tajima's D and nucleotide diversity could indicate a selective sweep for this particular TA system in CH populations, while in CO populations, the high nucleotide diversity and balancing selection suggest various variants of this TA system.

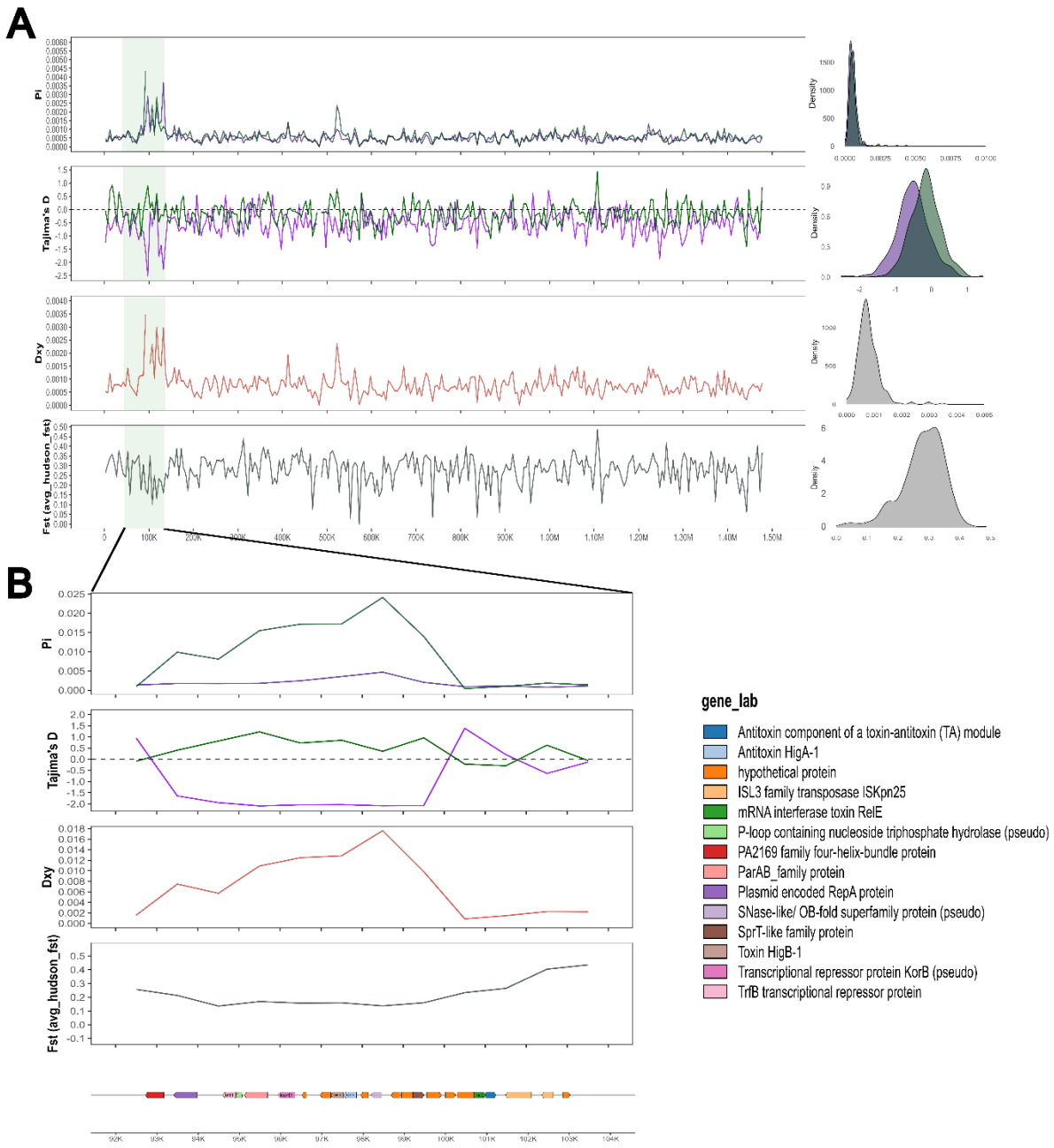


Figure 7. Population genomic statistics across the *Stammerula* MAG showing (A) from the top-down nucleotide diversity (π), Tajima's D, nucleotide divergence (D_{XY}), and population differentiation (F_{ST}) calculated over 5KB non-overlapping windows with their density plots on the right. B. Zoomed in on the 92-104KB region showing the same metrics but calculated across 1KB non-overlapping windows, with the Bakta annotated gene features plotted.

Discussion

In 1929, Hans-Jurgen Stammer described how unculturable bacterial symbionts lived in sheltered crypts along the midgut of tephritid flies (Stammer, 1929). Nearly a century later, those tenants were given a name, *Candidatus Stammerula spp.*, when Mazzon and colleagues (2008) used 16S rRNA analyses to show that each fly species carried its own distinct bacterial companion. In that study, 17 out of 25 tephritid species harbored a unique symbiont sequence, irrespective of where, when or from which host plant the fly was sampled. The exception was *Noeeta pupillata*, whose symbionts showed slight genetic variation correlating with the different hawkweed (*Hieracium*) host plant species the fly was sampled from (Mazzon et al. 2008). This exception hinted that the symbiont itself could potentially reflect the ecological niche of its host, raising the possibility that the population structure of bacterial symbionts could track the ecological divergence within a host species. Building on this foundation, my work extends this idea from a perhaps, to the finding that *Stammerula tephritidis* populations structure semi-accordingly to their host ecotype.

To investigate how *Stammerula* populations are structured requires its genome. By extracting out the microbial data from whole-metagenome samples, I assembled the nearly complete and highly contiguous MAG of *Stammerula* and compared the *Stammerula* populations across host ecotype and individually. I combined functional annotation with population genomics to investigate (1) the genes encoded on the MAG to infer biology and plausible functional role of *Stammerula*, (2) whether *Stammerula* exhibits congruent population structure with its host ecotype, (3) if divergence is associated with genomic regions or functions.

Functional role

The novel MAG of *Stammerula* that I assembled is the first draft genome for *Stammerula* which takes this crypt-dwelling symbiont into the genomic context. Importantly, its genomic features are characteristic of a host-restricted lifestyle. *Stammerula*'s genome is relatively small (1.48MB) and AT-biased (43% GC), reflecting a reduced, streamlined genome typical of insect symbiont that have shed many functions not required for the symbiotic environment (Fisher et al., 2017; Moran, 2006; Wernegreen, 2002). Functional annotation of the *Stammerula* MAG reveals a symbiont well-equipped to supplement its host's nutrition. The symbiont retains all pathways required for normal cellular function and aerobic respiration in the crypts, as well as fermentation under low oxygen as the Pta-Ack pathway is complete. Strikingly, *Stammerula* possesses an extensive biosynthetic repertoire for essential nutrients which are the amino acids and vitamins that the host can not synthesize itself (Serrato-Salas & Gendrin, 2022). KEGG pathway analysis showed complete pathways for the synthesis of B-vitamins riboflavin (B_2) and folate (B_9), as well as the genetic ability to produce 9 out of the 10 essential amino acids (arginine, histidine, isoleucine, leucine lysine, methionine, threonine, tryptophan, and valine (Chang, 2004)). Phenylalanine is the missing essential amino acid, and it is not complete due to the absence of the *tyrB* gene. Furthermore, *Stammerula* has the genetic capacity to produce a complete Gram-negative cell envelope, but the O-antigen biosynthesis is incomplete due to the key O-antigen ligase (*waaL*) missing. Genes for motility and constructing flagella are also missing, as well as any plant-degrading enzymes.

Taking these genomic findings into account strongly suggests that *Stammerula* is a nutrient provisioning symbiont for *T. conura*. The lack of cell wall degradation suggests that *Stammerula* metabolizes already catabolized sugars in the larval stage and potentially

provides a more important role during the adult stage of its host as adult *T. conura* flies feeds of nectar and other simple sugars. It is important to note, however, that this functional profile is only a hypothesis based on which genes are present. To truly understand the functional role of *Stammerula* requires transcriptomics to determine what and when the genes are expressed.

Symbiont population structure as a faint mirror reflection of host's ecological divergence

Despite *Stammerula*'s potential role, the symbiont was not universally present across all *T. conura* individuals (Figure 4). Yet, population genomic analyses of the 67 samples harboring *Stammerula* show that its population structure was semi-congruent with the host's ecological structure. *Stammerula* from flies using the same host plant tend to cluster together but it is not perfect as mixed CH+CO clades occur. Examples of groups clustering by transect and contact zone were found, indicating that the symbiont has experienced gene flow or sorting independently of the host lineages. Mazzon et al. (2010) found that *Stammerula* and its hosts have a weak cladogenesis and emphasized that it could be due to incomplete transmission or the potential exposure in the crypts. As *Stammerula* is an extracellular symbiont living inside crypts, and that various organism can cohabit the flower bud, occasional horizontal transfers between other flies and environment is possible (Mazzon et al., 2010). Possibly explaining the nonsignificant correlation of F_{ST} and geographic distance, and the tree topology. Moreover, bacteria can be lost or replaced, which would explain the Kraken2 results showing varying abundances and individual missingness.

The genome wide population genomic analysis of CH and CO samples therefore proved difficult as no clear loci could be determined. The binary division and higher mutation rates already make population genomics difficult, and with the added layer of metagenomics makes the analysis noisy. The effective population size nor the number of *Stammerula* strains present in one crypt or individual is not known, proving this task difficult. One clear finding however, was that *Stammerula* lineages in the sympatric Skåne populations (CHSK and COSK) are not being horizontally transferred showing strong support for the absence of gene flow. Nonetheless, the population structure and phylogeny do show a reflection of the host ecotype divergence as both the host tree and symbiont tree indicate that CO ecotype is the derived state, regardless of the occasional host-switching bacteria (Figure 7, Supplementary Figure 4).

Looking forward, future research can take several directions. As a high-quality MAG of *Stammerula* exists, metatranscriptomics is a potential path to truly understanding the symbiont's function. Moreover, investigating the fitness of flies with or not with *Stammerula* across life stages could be informative to determine if *Stammerula* is a facultative or an obligate symbiont. Not reported here, but high-quality MAGs from each population were assembled through my pipeline, where a pangenome analysis could shed light on core genes across all populations, and the genes distinguishing transect, contact zone, and host race.

Code and Data Availability

Pipeline for metagenomics is available here: https://github.com/ndreey/stammerula2025
Analysis code is found here: https://github.com/ndreey/population_metagenomics

Long and short read sequences have been archived on NCBI SRA (https://dataview.ncbi.nlm.nih.gov/object/PRJNA1160207) and will be held under embargo until publication.

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From a podcast about the microbiome on the palms of our hands to writing my thesis on the microbes inside flies. I have learned to find wonder in everything, even the most mundane.

Thank you!

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